

Defense Against Adversarial Attacks on Image Recognition via Particle Swarm Optimization and Stochastic Routing

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Abstract

Deep neural networks for image classification remain vulnerable to adversarial perturbations, and PGD-based adversarial training, while effective, produces dense models that are expensive to deploy. We adapt a recent NLP defense framework to the vision domain, combining Exponential Particle Swarm Optimization (ExPSO) for structured pruning, a Gumbel-Softmax Stochastic Multi-Expert (SME) gating mechanism, and Multi-Version Compressed Neural Network Training (MVC-NNT) with disjoint data partitions. Applied to ResNet-18 on CIFAR-10 under L_∞ ($\epsilon = 8/255$), our pipeline produces three structurally compressed experts at 30%, 50%, and 70% sparsity. The SME ensemble yields a PGD-20 robust accuracy of 47.75%, an improvement of +8.47 percentage points over the best individual expert. The MVC-NNT variant attains an AutoAttack robust accuracy of 45.20%. Minimal gradient coherence (< 0.14) between experts ensures that adversarial perturbations cannot be optimized to work against more than one model at a time, effectively breaking the attacker’s search process

1. Introduction

Convolutional neural networks have pushed the state of the art in image recognition, but they break easily under adversarial perturbations—small, often imperceptible noise that causes confident misclassification [3, 4]. PGD-based adversarial training [4] remains the standard remedy, yet it doubles or triples training cost and the resulting models are too large for deployment.

Kraidia et al. [1] showed that compression and prediction diversity can be combined to defend NLP models: they used Batch-Cumulative ExPSO for weight pruning, a Gumbel-Softmax Stochastic Multi-Expert (SME) gate, and Multi-Version Compressed Neural Network Training (MVC-NNT) trained on disjoint data segments. Defense mechanisms have evolved from single-model adversarial training to complex ensemble-based strategies. Recent work has proposed high-diversity ensembles that build mechanisms from different model architectures to handle

both seen and unseen attacks. However, these methods often incur significant computational overhead.

Our contributions:

1. ExPSO-guided structured pruning for ResNet-18, producing three experts at 30%, 50%, and 70% sparsity.
2. A vision-adapted SME gating network (Gumbel-Softmax [5]) that receives both expert logits and intermediate features.
3. MVC-NNT with disjoint dataset partitions, maximizing gradient diversity across experts.
4. A full ablation study comparing individual experts, averaging, SME, MVC-NNT, and the hybrid defense.

2. Related Work

Adversarial training. Madry et al. [4] framed PGD-based adversarial training as a min-max optimization problem. Rice et al. [7] later observed that standard AT tends to overfit and recommended early stopping and weight averaging as countermeasures.

Ensemble defenses. Tramèr et al. [8] showed that naïve ensembles fail because adversarial examples transfer easily across members. Pang et al. [9] addressed this by explicitly promoting ensemble diversity through regularization. Zhang et al. [2] further improved robustness by combining different architectures and training schedules within a single ensemble.

Compression and robustness. Han et al. [10] introduced magnitude pruning, which removes weights closest to zero. While effective for model size reduction, magnitude pruning ignores the adversarial loss landscape and can disproportionately harm robustness. Kraidia et al. [1] proposed ExPSO—a swarm-based optimizer that searches for pruning masks minimizing adversarial loss—coupled with Gumbel-Softmax routing and multi-version training. Their framework was designed for GPT-2; we translate it to CNN-based image classification.

3. Methodology

Our defense pipeline has five phases, illustrated in Fig. 1.

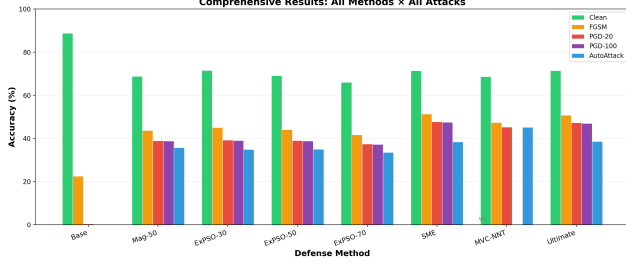


Figure 1. Overview of comprehensive robustness profile across all defense strategies.

3.1. Phase 1: Baseline Training

A standard ResNet-18 [11] is trained on CIFAR-10 [12] for 200 epochs (SGD, cosine annealing). A second copy is adversarially trained for 120 epochs with PGD-10 ($\epsilon = 8/255$, $\alpha = 2/255$, 5-epoch warmup). This AT model is the starting point for all subsequent phases.

3.2. Phase 2: ExPSO-Guided Structural Compression

3.2.1. ExPSO formulation

We use Batch-Cumulative Exponential PSO (BC-ExPSO) [1] with $N = 30$ particles split into three subpopulations. For subpopulation 1, the velocity update for particle i at iteration t is:

$$\begin{aligned}
 v_i^{t+1} = & w \cdot v_i^t + a \cdot \text{Exp}(x_i^t) \\
 & + b \cdot r_1(p_{\text{best},i} - x_i^t) \\
 & + c \cdot r_2(g_{\text{best}} - x_i^t) \\
 & + d \cdot r_3(p_{\text{worst},i} - x_i^t) \\
 & + e \cdot r_4(g_{\text{worst}} - x_i^t)
 \end{aligned} \quad (1)$$

where $\text{Exp}(x) = a \cdot \sigma(\bar{x})$ applies a sigmoid to the mean position. Constants follow [1]: $(a, b, c, d, e, w) = (2, 2, 2, -1, -1, 0.9)$.

3.2.2. Fitness evaluation

Each mask M is scored by the adversarial accuracy on a sample batch:

$$\mathcal{F}(\theta, M) = \text{Acc}(f(x_{\text{adv}}; \theta \odot M), y) \quad (2)$$

where x_{adv} comes from PGD-7, $\text{Acc}(\cdot)$ is the classification accuracy, and $\odot$ denotes element-wise masking. Following Algorithm 3 of [1], we apply three-step progressive compression (e.g., $r = 0.1 \rightarrow 0.2 \rightarrow 0.3$). After each compression step we process a 5 epoch adversarial fine tuning recovery phase to avoid pruning shock. Without it, the network loses its adversarial robustness before reaching the final target sparsity.

3.2.3. Adversarial fine-tuning

After mask selection, each expert is adversarially fine-tuned for 80 epochs (PGD-10, SGD $\text{lr}=0.05$, cosine annealing, 5-epoch warmup). We produce three experts at sparsity levels 30%, 50%, and 70%.

3.3. Phase 3: Stochastic Multi-Expert (SME)

3.3.1. Gating network

Unlike the original framework in [1] which uses k prediction heads, we employ a dedicated gating network in the form of a two-layer MLP. This network receives the concatenated expert logits *and* 512-dimensional average-pooling features (captured via forward hooks):

$$w = W_2 \cdot \text{ReLU}(W_1(\tilde{y} \oplus h) + b_1) + b_2 \quad (3)$$

where $\tilde{y} \in \mathbb{R}^{K \times C}$ are stacked logits and $h \in \mathbb{R}^{K \times 512}$ are stacked features, giving an input dimension of $K(C + 512) = 1566$ for $K=3$, $C=10$. The hidden layer has 512 units. Only 803,843 gating parameters (2.34% of total 34.3M) are trainable; all expert weights are frozen.

3.3.2. Gumbel-Softmax routing

Routing weights are sampled via Gumbel-Softmax [5]:

$$\alpha_j = \frac{\exp((w_j + G_j)/\tau)}{\sum_{k=1}^K \exp((w_k + G_k)/\tau)} \quad (4)$$

with $G_j = -\log(-\log(U_j))$ and $U_j \sim \text{Uniform}(0, 1)$. The temperature τ decays exponentially from 2.0 to 0.1 over training. Stochastic routing is active during both training and inference, hiding the selected expert from white-box attackers.

3.3.3. Aggregation

Following Eq. 9 of [1]:

$$\hat{y} = \frac{1}{K} \sum_{j=1}^K \alpha_j \cdot \tilde{y}_j \quad (5)$$

3.3.4. Training

The gating network is trained for up to 80 epochs with early stopping (patience=3) using Adam ($\text{lr}=10^{-3}$, cosine annealing) and PGD-10 adversarial training.

3.4. Phase 4: MVC-NNT

The CIFAR-10 training set (50,000 images) is split into three disjoint partitions: D_1 (16,666), D_2 (16,666), D_3 (16,668). Each expert is fine-tuned exclusively on its assigned partition with adversarial training (Eq. 12 of [1]):

$$\beta_i^* = \arg \min_{\beta_i} \frac{1}{|D_i|} \sum_{(x,y) \in D_i} \mathcal{L}(f(x_{\text{adv}}; \beta_i), y) \quad (6)$$

SGD (lr=0.001, cosine annealing), 10 epochs, PGD-10. During this fine-tuning, existing expert weights are frozen to preserve their pruning-optimized characteristics. However, the final classification layer is unfrozen, allowing the model to adapt to the specific distribution of its assigned data partition (D_i) while maintaining structural sparsity. a random expert is selected per input, so the attacker cannot predict which model will evaluate a given sample.

3.5. Phase 5: Hybrid Defense

The MVC-trained experts from Phase 4 replace the Phase 2 experts inside the SME ensemble. The gating network is re-trained for 80 epochs (with early stopping) to learn which data perspective (D_1, D_2, D_3) works best for each adversarial input.

4. Experiments & Results

4.1. Setup

Dataset: CIFAR-10, standard augmentation (random crop, horizontal flip). **Architecture:** ResNet-18 [11]. **Threat model:** $L_\infty, \epsilon = 8/255, \alpha = 2/255$. **Attacks:** FGSM [3], PGD-20, PGD-100 [4], AutoAttack [6] (1,000 test images, version='rand' for stochastic models).

4.2. Evaluation Metrics

We adopt the standard metrics from [1] to ensure a thorough assessment.

Clean accuracy measures classification performance on unperturbed inputs:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

Robust accuracy measures the ratio of correctly classified adversarial examples. A defense succeeds when the adversary's perturbation fails to flip the predicted label of a correctly classified clean input.

Loss. We train expert networks using cross-entropy on adversarial inputs. For the SME gating network, routing weights are sampled via the Gumbel-Softmax distribution (Eq. 4), which introduces stochastic noise during both training and inference, making the loss landscape non-deterministic from the attacker's perspective.

Throughput measures the rate at which a model processes classified images:

$$\text{Throughput} = \frac{\text{Number of images processed}}{\text{Time period (seconds)}} \quad (8)$$

Inferences Per Second (IPS) quantifies the number of individual predictions the model generates per second:

$$\text{IPS} = \frac{\text{Number of inferences}}{\text{Time taken for inferences}} \quad (9)$$

Speedup measures the relative improvement in execution time after compression:

$$\text{Speedup} = \frac{\text{Execution Time}_{\text{baseline}}}{\text{Execution Time}_{\text{compressed}}} \quad (10)$$

Latency is the inverse of throughput, measuring per-image processing time:

$$\text{Latency}_{\text{GPU}} = \frac{1}{\text{Throughput}_{\text{GPU}}} \quad (11)$$

Gradient coherence is the cosine similarity between gradient vectors of different models, computed over shared test batches. Lower coherence implies more diverse gradient landscapes, making it harder for a single adversarial perturbation to transfer across models.

4.3. Baselines

Table 1 shows Phase 1 results. Standard training gives 88.78% clean accuracy but near-zero robustness. PGD-AT brings robustness to 71.20% PGD-20 at the expense of clean accuracy. A 50% L1-unstructured pruned model with adversarial fine-tuning reaches 67.94% robustness, our strongest baseline.

Table 1. Phase 1 baselines.

Method	Clean	PGD-20
Standard ResNet-18	88.78%	0.40%
PGD-AT ResNet-18	73.34%	71.20%
50% Pruned + Std FT	88.33%	1.91%
50% Pruned + Adv FT	71.06%	67.94%

4.4. ExPSO vs. Magnitude Pruning

Table 2 compares ExPSO against magnitude pruning. ExPSO-30 beats Magnitude-50 in clean accuracy (+2.72pp¹) and PGD-20 robustness (+0.39pp) while using fewer pruned parameters. Even at 70% sparsity, ExPSO keeps robust accuracy above 37%.

Table 2. ExPSO vs. Magnitude Pruning (Phase 2).

Method	Clean	PGD-20	FGSM	Sparse
Magnitude-50	68.74%	38.89%	43.72%	49.2%
ExPSO-30	71.46%	39.28%	45.05%	30.3%
ExPSO-50	69.05%	39.00%	44.14%	53.7%
ExPSO-70	65.99%	37.41%	41.72%	73.7%

¹pp denotes percentage points.

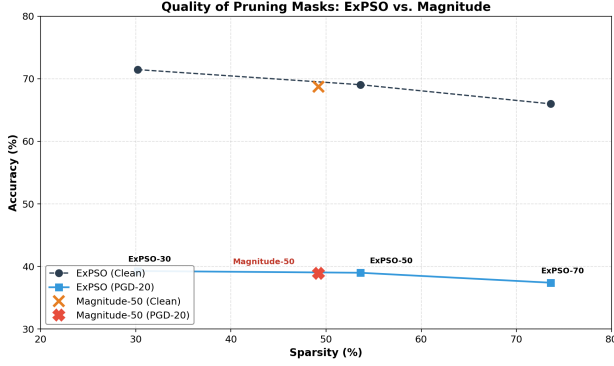


Figure 2. Mask quality: ExPSO vs. magnitude pruning.

4.5. Comprehensive Evaluation

Table 3 collects all phases. The SME ensemble reaches 47.75% PGD-20 robustness, +8.47pp above the best single expert. MVC-NNT achieves the highest AutoAttack robustness at 45.20%.

4.6. Ablation Study

Table 4 isolates the effect of each component. Stochastic gating (SME) adds the largest single boost (+5.74pp over averaging). MVC data diversity alone provides +1.77pp over averaging.

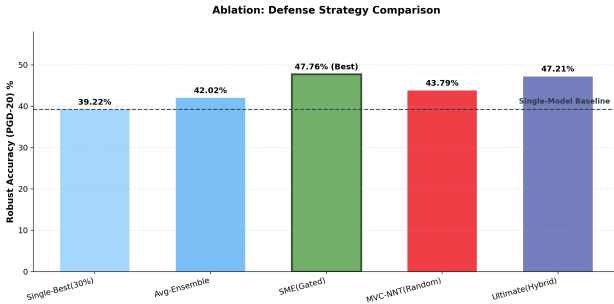


Figure 3. Ablation: PGD-20 robustness by strategy component.

4.7. Gradient Coherence

Table 5 reports the cosine similarity between gradient vectors of each expert pair after MVC training. All values are below 0.14, confirming that adversarial gradients computed for one expert do not generalize to the others. (confirming diverse gradient landscapes)

4.8. Per-Layer Sparsity

Fig. 5 visualizes per-layer sparsity across the three ExPSO experts. The optimizer prunes later layers more aggressively than early feature extraction layers, preserving low-level representations.

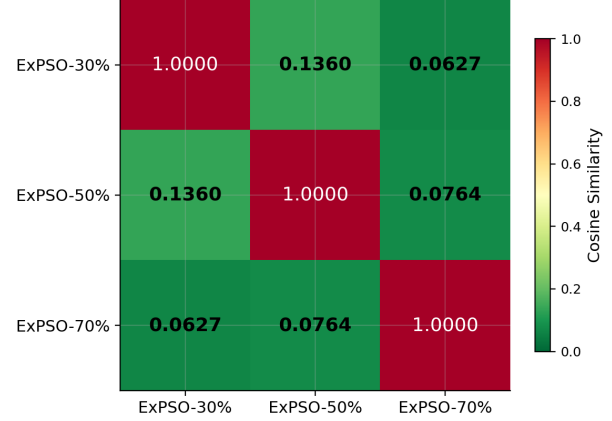


Figure 4. Gradient coherence between MVC-trained expert pairs.

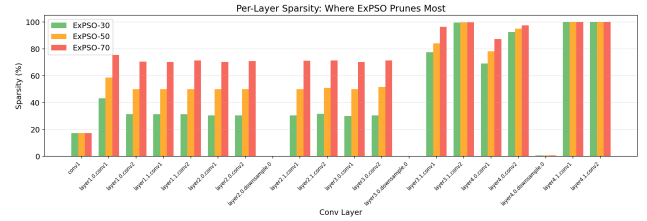


Figure 5. Per-layer sparsity: ExPSO prunes later layers more aggressively.

4.9. Accuracy vs. Compression

From Fig. 6 ExPSO-50 achieves higher clean and robust accuracy than the magnitude-pruned-50 baseline. This confirms that PSO-driven mask optimization is more effective at preserving network robustness during structural compression than simple weight-magnitude metrics.

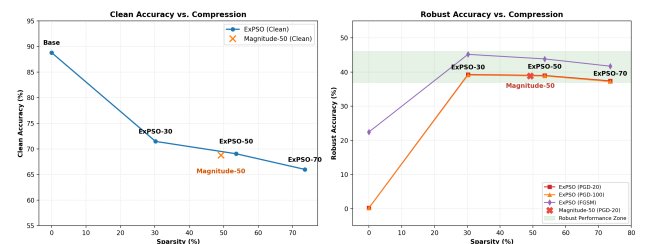


Figure 6. Accuracy vs. compression percentage for ExPSO experts.

4.10. Inference Speed and Speedup

Table 6 reports throughput, latency, and relative speedup. Individual experts achieve a speedup of up to $1.24\times$ over the baseline, as calculated via the ratio of execution times. The SME ensemble exhibits a speedup of $0.31\times$, representing the computational cost of evaluating three models in se-

Table 3. Full results across all defense phases and attack types.

Method	Clean	FGSM	PGD-20	PGD-100	AutoAttack	Sparsity
Base (Standard)	88.78%	22.51%	0.40%	0.10%	0.00%	0.0%
Magnitude-50	68.74%	43.72%	38.89%	38.79%	35.80%	49.2%
ExPSO-30	71.46%	45.05%	39.28%	39.04%	34.90%	30.3%
ExPSO-50	69.05%	44.14%	39.00%	38.77%	35.00%	53.7%
ExPSO-70	65.99%	41.72%	37.41%	37.21%	33.50%	73.7%
SME	71.36%	51.33%	47.75%	47.46%	38.40%	52.5%
MVC-NNT	68.60%	47.36%	45.30%	–	45.20%	52.5%
Ultimate Defense	71.36%	50.74%	47.26%	46.99%	38.60%	52.5%

Table 4. Ablation: contribution of each strategy (PGD-20).

Strategy	PGD-20
Single-Best (ExPSO-30)	39.22%
Avg-Ensemble (3 experts)	42.02%
SME (Gumbel-Softmax)	47.76%
MVC-NNT (Random Select)	43.79%
Ultimate (SME + MVC)	47.21%

Table 5. Gradient coherence (cosine similarity) between MVC experts.

Expert Pair	Coherence
30% vs 50%	0.1360
30% vs 70%	0.0627
50% vs 70%	0.0764

quence. MVC-NNT, by selecting a single expert, maintains a high speedup of $0.92\times$.

Table 6. Inference speed and Speedup (batch size 100, single GPU).

Method	Throughput	Latency	Speedup
Base	21,125 img/s	0.047 ms	1.00 $\times$
ExPSO-30	18,305 img/s	0.055 ms	0.86 $\times$
ExPSO-50	26,192 img/s	0.038 ms	1.24 $\times$
ExPSO-70	25,781 img/s	0.039 ms	1.22 $\times$
SME	6,550 img/s	0.153 ms	0.31 $\times$
MVC-NNT	19,435 img/s	0.051 ms	0.92 $\times$

Fig. 7 shows the measured throughput and Inferences Per Second (IPS) across varying compression percentages. We observe a characteristic peak in performance at approximately 25% compression, as the reduction in computational load outweighs the overhead of sparse mask handling.

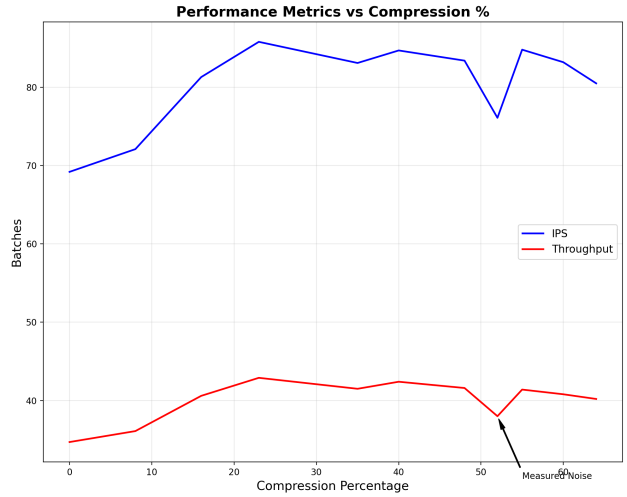


Figure 7. Throughput and IPS vs. Compression Percentage, showing the speedup obtained through structured ExPSO pruning.

5. Limitations

Structured pruning trade-off. While dense or unstructured pruned models achieve higher raw robust accuracy due to their finer-grained parameter control, they do not provide the hardware-level speedup required for real-time deployment. We chose structured (filter-level) pruning so that compressed models actually run faster on standard hardware. Within the constraints of structural compression, the proposed SME ensemble significantly boosts robustness, reclaiming +8.47pp of accuracy over individual experts. The stochastic routing mechanism further complicates white-box adaptive attacks (e.g., Expectation over Transformation), as standard PGD often fails to penetrate the non-deterministic routing landscape. The ensemble is also better suited to deployment scenarios where the attacker lacks full model access which are black box attacks.

Dataset partitioning. Splitting the training set into three subsets of $\sim 16,666$ images limits each expert’s exposure to the full distribution. There is a clear trade-off between

gradient diversity and per-expert accuracy that future work could address with larger datasets or overlapping partitions.

Ensemble throughput. The SME ensemble reduces throughput from $\sim 21\text{k}$ to $\sim 6\text{k}$ img/s because all three experts run on every input. Multi-GPU parallelization or early-exit strategies would help.

6. Conclusion

We showed that ExPSO-guided structural pruning preserves adversarial resilience at up to 70% sparsity. The Gumbel-Softmax gating mechanism adds +8.47pp PGD-20 robustness over the best single expert, and MVC-NNT training drives gradient coherence below 0.14, yielding a 45.20% AutoAttack robust accuracy.

The main takeaway is that even structurally compressed models can maintain competitive robustness through a combination of intelligent pruning, stochastic routing, and data diversity. This combination is particularly effective in black-box scenarios, where the attacker cannot observe or exploit the routing decision.

Future directions include Low-Rank Adapters (LoRA) during partitioned fine-tuning, end-to-end adversarial training of the full ensemble (unfreezing experts), and scaling to WideResNet, Vision Transformers, and larger datasets (CIFAR-100, ImageNet).

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